

ONLINE RETAIL ANALYTICS

Executive Summary

UCI Online Retail Dataset | December 2010 - December 2011

397,884

Clean Transactions

4,338

Unique Customers

GBP 480.87

Avg Order Value

93%

Repeat Revenue Share

1. Project Overview

This project delivers a full end-to-end analysis of transactional data from a UK-based online gift retailer. The dataset covers December 2010 through December 2011 and contains 541,909 raw transactions across 38 countries. After rigorous data cleaning, 397,884 records were retained for analysis.

The analysis was conducted in three phases, progressing from descriptive reporting through deeper customer and revenue analysis to data-backed business recommendations. All findings are visualised across a three-page interactive Tableau dashboard.

Important data quality note: The 2010 data is incomplete, with several months containing only a single day of transactions. All trend analysis and year-over-year comparisons in this report are based on 2011 data only.

2. Data Preparation

The raw dataset required significant cleaning before analysis could begin:

- Removed 135,080 rows with missing CustomerID values
- Removed 1,454 rows with missing product descriptions
- Removed 9,288 cancelled transactions (InvoiceNo prefix C)
- Removed rows where Quantity or UnitPrice were zero or negative
- Converted InvoiceDate from string to datetime format
- Converted CustomerID from float to string identifier
- Engineered a Revenue column (Quantity multiplied by UnitPrice)

Final cleaned dataset: 397,884 rows, 9 columns, zero null values.

3. Key Findings

3.1 Sales Trends

- Revenue grew steadily through 2011, peaking in September at GBP 1.09M and again in November at GBP 1.04M, suggesting a strong pre-holiday gifting season.
- New customer acquisition declined from 544 in January 2011 to 230 by November 2011 -- a 58% drop over the year. This is a significant concern for long-term growth.
- Cohort retention improved consistently through 2011, with later cohorts showing stronger month-on-month retention rates, indicating the business is getting better at keeping existing customers even as acquisition slows.

3.2 Customer Behaviour

- 93% of total revenue comes from repeat customers. The business runs almost entirely on loyalty rather than new customer acquisition.
- 74.6% of revenue is generated by just 20% of customers, representing a significant concentration risk.
- RFM segmentation across 4,338 customers identified: 712 Champions (recent, frequent, high spending), 1,216 At Risk (inactive 2-6 months), 958 Lost (inactive over 6 months), and 28 Loyal customers.
- The average order value is GBP 480.87, reflecting the wholesale nature of the customer base where buyers are predominantly small businesses purchasing in bulk.

3.3 Product Performance

- Paper Craft Little Birdie generated GBP 168,470 from a single customer -- a single-point dependency risk that should be flagged to senior management.
- Regency Cakestand 3 Tier and Rabbit Night Light are the most consistent performers across international markets, appearing in top product lists for multiple countries.
- Administrative entries (Postage, Manual, Discount) appear throughout product revenue data and distort product-level reporting. These require system-level tagging to separate from genuine product sales.
- Top 20% of customers buy products such as Chilli Lights and Medium Ceramic Top Storage Jar that the bottom 80% do not -- potential upsell gateway products.

3.4 Geographic Markets

- The United Kingdom accounts for 89% of all transactions, leaving international markets significantly underdeveloped relative to their potential.
- Top international markets by revenue: Netherlands (GBP 285,446), Ireland (GBP 265,546), Germany (GBP 228,867), France (GBP 209,024), and Australia (GBP 138,521).

- Singapore and Netherlands have the highest average order values at over GBP 3,000 per order, approximately seven times the UK average, indicating large wholesale buyers in these markets.
- Significant country product opportunity gaps were identified -- for example, 3 Tier Cake Tin Red generates GBP 4,236 in Ireland but zero sales in Netherlands, and 5 Hook Hanger Red Magic Toadstool generates GBP 2,923 in Netherlands but zero in Australia and Ireland.

4. Data-Backed Recommendations

The following recommendations are derived directly from the data analysis. Each is supported by specific findings and includes a proposed next step for implementation.

Recommendation	Data Insight	Next Step
1. Win-Back At Risk Customers	1,216 customers inactive for 2-6 months. Their last purchases were Regency Cakestand, Party Bunting, Jumbo Bag Red Retrospot and White Hanging Heart T-Light Holder.	Launch a targeted email campaign anchored to these specific products. Offer a 10% discount with a clear expiry date to create urgency before customers move into the Lost segment.
2. Protect Champion Customers	712 Champion customers drive the bulk of the 93% repeat revenue. With 74.6% of revenue in the top 20% of customers, losing even a small number would have a disproportionate impact.	Introduce dedicated account management, early access to new products, or a formal loyalty programme. Assign relationship owners to the top 50 accounts by spend immediately.
3. Convert One-Time Buyers	1,493 customers made only one purchase. Their most common first purchases were Regency Cakestand, White Hanging Heart and Party Bunting -- the same gateway products bought by Champions.	Trigger a follow-up email 30 days after first purchase featuring complementary products. Customers who make a second purchase are significantly more likely to become long-term repeat buyers.
4. Exploit Country Product Gaps	Multiple products have strong sales in one country but zero in others. 3 Tier Cake Tin Red: GBP 4,236 in Ireland, zero in Netherlands. 5 Hook Hanger: GBP 2,923 in Netherlands, zero in Australia.	Run country-specific email or social campaigns featuring these products in underperforming markets. Low risk since demand is already proven elsewhere.
5. Rebuild New Customer Acquisition	Acquisition dropped 58% from January to November 2011 (544 to 230 new customers per month). Without growth in new customers, even strong retention will eventually shrink the customer base.	Review all active marketing and acquisition channels. Set a monthly new customer target and measure against it. Consider referral incentives targeting the existing Champion segment.
6. Reduce Revenue Concentration Risk	74.6% of revenue from 20% of customers. Paper Craft Little Birdie	Promote top 20% exclusive products (Chilli Lights, Medium

	generates GBP 168,470 from a single buyer. Losing key accounts would have an outsized revenue impact.	Ceramic Top Storage Jar) to broader customer segments via targeted campaigns. Flag single-customer product dependencies to senior leadership.
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5. Dashboard Overview

All findings are presented across three interactive Tableau dashboard pages:

Page 1 -- Business Overview

Monthly revenue trend, top 10 products by revenue, repeat vs one-time customer revenue split, top 10 international markets, average order value by country, top 10 customers by spend.

Page 2 -- Customer and Revenue Insights

New customer acquisition trend, cohort retention heatmap, RFM customer segmentation, revenue concentration (80/20 analysis), top products by country, high revenue single-customer products.

Page 3 -- Data-Backed Recommendations

Product overlap between top 20% and bottom 80% customers, country product opportunity gaps heatmap, At Risk customer last purchases, one-time customer purchase patterns.

6. Tools and Methodology

Tool	Purpose
Python (Google Colab)	Data cleaning, exploratory analysis, customer segmentation, cohort retention, recommendation analysis
Pandas	Data manipulation, groupby operations, merges, filtering, aggregations
Tableau Public	Interactive dashboard visualisation across three pages
GitHub	Version control and portfolio hosting
UCI Online Retail Dataset	Source data -- publicly available transactional dataset from a UK-based gift retailer

Dataset source: Dr. Daqing Chen, School of Engineering, London South Bank University